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EXPERIMENT 2

Zero-shot, One-shot and Few-shot Prompts - 200-word Blog on "Applications of Artificial Intelligence in Healthcare"

AIM

To design zero-shot, one-shot and few-shot prompts for generating a 200-word blog on "Applications of Artificial Intelligence in Healthcare" and to compare the outputs.

PROCEDURE

1. Open an LLM chat interface (such as ChatGPT, Gemini or Claude) in a new chat.
2. Zero-shot: enter an instruction that describes only the task (write a 200-word blog on the given topic) without any example, and note the response.
3. One-shot: start a new chat and enter the same instruction together with ONE worked example, followed by the actual task. Note the response.
4. Few-shot: start a new chat and enter the same instruction together with TWO worked examples, followed by the actual task. Note the response.
5. Capture a screenshot of each prompt with its generated output.
6. Compare the three outputs and record the observations.

PROMPTS (CODE)

Zero-shot prompt:

```
Write a 200-word blog on "Applications of Artificial Intelligence in Healthcare."
```

One-shot prompt:

```
Write a 200-word blog in the same structure as the example (title, intro, three
numbered applications, conclusion).

Example:
Title: AI in Education - Learning That Adapts
Intro: Artificial Intelligence is changing how students learn and teachers teach.
1. Personalised learning: Adaptive platforms adjust lessons to each student's
pace.
2. Smart assessment: Automated grading gives instant feedback and saves teachers
time.
3. Virtual tutors: Chatbots answer doubts at any hour.
Conclusion: Used wisely, AI supports teachers and helps every learner succeed.

Now write a 200-word blog on: "Applications of Artificial Intelligence in
Healthcare."
```

Few-shot prompt:

```
Write a 200-word blog following the structure of the examples (title, intro,
three numbered applications, conclusion).
```

```
Example 1:
Title: AI in Education - Learning That Adapts
Intro: Artificial Intelligence is changing how students learn and teachers teach.
1. Personalised learning: Adaptive platforms adjust lessons to each student's
pace.
2. Smart assessment: Automated grading gives instant feedback and saves teachers
time.
3. Virtual tutors: Chatbots answer doubts at any hour.
Conclusion: Used wisely, AI supports teachers and helps every learner succeed.

Example 2:
Title: AI in Agriculture - Smarter Farming
Intro: Farmers are turning to Artificial Intelligence to grow more with less.
1. Crop monitoring: Drones and image analysis spot disease early.
2. Yield prediction: Models forecast harvests from weather and soil data.
3. Precision irrigation: Sensors water crops only when needed.
Conclusion: AI helps farmers save resources and secure food for the future.

Now write a 200-word blog on: "Applications of Artificial Intelligence in
Healthcare."
```

OUTPUT

Exp 2 - Zero-shot prompting (sample output)

PROMPT

Write a 200-word blog on "Applications of Artificial Intelligence in Healthcare."

MODEL OUTPUT

Applications of Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming healthcare by helping doctors diagnose diseases faster, treat patients more accurately and manage hospitals more efficiently. One of its most important uses is medical imaging. AI models can study X-rays, CT scans and MRIs to detect cancer, fractures and eye diseases, often spotting patterns that are hard for the human eye to see.

AI also speeds up drug discovery. Instead of testing millions of compounds in a laboratory, researchers use machine learning to predict which molecules are most likely to work, saving years of effort and large amounts of money. In patient care, virtual health assistants answer common questions, remind patients to take medicines and book appointments at any time of day.

Wearable devices powered by AI monitor heart rate, sleep and oxygen levels and warn users about possible problems early. Hospitals use predictive analytics to forecast admissions, manage staff and reduce waiting times. Robotic surgery systems guided by AI help surgeons perform delicate procedures with greater precision.

Although AI raises concerns about data privacy and bias, its benefits are significant. With careful regulation and human supervision, AI can make healthcare more accessible and effective.

Figure: Zero-shot prompt and sample blog

Exp 2 - One-shot prompting (sample output)

PROMPT

Write a 200-word blog in the same structure as the example (title, intro, three numbered applications, conclusion).

Example:
Title: AI in Education - Learning That Adapts
Intro: Artificial Intelligence is changing how students learn and teachers teach.
1. Personalised learning: Adaptive platforms adjust lessons to each student's pace.
2. Smart assessment: Automated grading gives instant feedback and saves teachers time.
3. Virtual tutors: Chatbots answer doubts at any hour.
Conclusion: Used wisely, AI supports teachers and helps every learner succeed.

Now write a 200-word blog on: "Applications of Artificial Intelligence in Healthcare."

MODEL OUTPUT

Title: AI in Healthcare - Smarter Care for Everyone

Healthcare is changing quickly, and Artificial Intelligence is leading the way. From the clinic to the laboratory, smart algorithms help professionals work faster and care better. Here are three ways it is making a difference today.

1. Early diagnosis: AI systems analyse X-rays, MRIs and CT scans within seconds and flag signs of cancer, stroke or eye disease that might be missed. Radiologists then confirm the findings, saving valuable time. Earlier detection means simpler treatment and better recovery.

2. Drug discovery: Developing a new medicine can take over a decade. Machine learning predicts how molecules behave, helping scientists shortlist promising candidates and cut time and cost. This could bring life-saving drugs to patients years sooner.

3. Personalised treatment: By studying a patient's history, genetics and lifestyle, AI suggests therapies suited to that individual instead of one-size-fits-all methods. This improves outcomes and reduces side effects.

Conclusion: Challenges remain. Patient data must be protected, algorithms must be tested for bias, and doctors should always make the final decision. Used responsibly, AI will not replace healthcare workers; it will give them more time to care for people. Their expertise, combined with AI, is the best path forward.

Figure: One-shot prompt and sample blog

Exp 2 - Few-shot prompting (sample output)

PROMPT

Write a 200-word blog following the structure of the examples (title, intro, three numbered applications, conclusion).

Example 1:
Title: AI in Education - Learning That Adapts
Intro: Artificial Intelligence is changing how students learn and teachers teach.
1. Personalised learning: Adaptive platforms adjust lessons to each student's pace.
2. Smart assessment: Automated grading gives instant feedback and saves teachers time.
3. Virtual tutors: Chatbots answer doubts at any hour.
Conclusion: Used wisely, AI supports teachers and helps every learner succeed.

Example 2:
Title: AI in Agriculture - Smarter Farming
Intro: Farmers are turning to Artificial Intelligence to grow more with less.
1. Crop monitoring: Drones and image analysis spot disease early.
2. Yield prediction: Models forecast harvests from weather and soil data.
3. Precision irrigation: Sensors water crops only when needed.
Conclusion: AI helps farmers save resources and secure food for the future.

Now write a 200-word blog on: "Applications of Artificial Intelligence in Healthcare."

MODEL OUTPUT

Title: AI in Healthcare - Care That Thinks Ahead

Intro: Hospitals and clinics are adopting Artificial Intelligence to detect illness earlier, treat patients better and run services smoothly. These tools do not replace experts; they extend what experts can see and do.

1. Medical imaging: Deep learning models read X-rays, CT scans and MRIs in seconds, highlighting tumours, fractures and eye disease. Doctors receive a reliable second opinion, which supports faster and more accurate diagnosis. Hospitals with limited specialists benefit the most.

2. Drug discovery: Traditional drug development is slow and expensive. AI screens millions of molecules virtually and predicts which will work, helping new medicines reach patients sooner. This approach also reduces laboratory costs and failed trials.

3. Virtual health assistants and wearables: Chatbots answer questions, send medicine reminders and book appointments, while smartwatches track heart rate and oxygen levels and alert users to early warning signs. Doctors can then step in before a minor issue becomes serious.

Conclusion: AI cannot replace the empathy and judgement of healthcare professionals, but it gives them powerful support. With strong privacy protection and human oversight, AI can make quality healthcare accessible to everyone. Together, humans and machines can deliver safer, faster and fairer care.

Figure: Few-shot prompt and sample blog

RESULT

The three outputs contained 196, 203 and 202 words respectively, all close to the 200-word target. The zero-shot blog was written as flowing paragraphs, whereas the one-shot and few-shot blogs followed the title, introduction, three numbered applications and conclusion structure given in the examples, which made them easier to scan.

CONCLUSION

All three prompting styles produced relevant blogs of about 200 words, but examples control the structure. One-shot and few-shot prompts gave a predictable layout, so they are preferred when a fixed blog format is required.